STAFF REVIEW · H446 195 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. $0.1000_2 \times 2^3$.

2. 11111111.

3. 8.

4. 0.1100_2 = decimal 0.75.

I Do — source worked steps

1. Align — B=0.1000×2^2 becomes 0.0100×2^3.

2. Subtract — $0.1100 - 0.0100 = 0.1000$.

3. Check — $0.1000_2 \times 2^3 = 4$; this matches decimal $6 - 2$.

We Do — guided CHECK

1. Add and check
$0.1010 + 0.0010 = 0.1100 \times 2^3$ = decimal 6.

You Do — source mark schemes

1. So corresponding mantissa bits represent the same place values.

2. Align B to 0.0100×2^3. Subtract: $0.1010 - 0.0100 = 0.0110 \times 2^3$. Shift left and decrease exponent: $0.1100 \times 2^2 = 3$. No shown bits are discarded.

3. Right-shifting may discard significant low-order bits. They cannot all be held in the available mantissa, so rounding or truncation can produce an inexact result.

Exit Ticket — source answer

1. True mathematically; finite-width storage may discard significant bits.